Task 1

① $\beta = \{(1,0,0),(0,1,0),(0,0,1)\}$

$T(1,0,0) = (1,0)$

$T(0,1,0) = (-1,0)$

$T(0,0,1) = (0,2)$

$[T]_\beta^\gamma = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ 訂正

$[T]_\beta^\gamma = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$T: \mathbb{R}^3 \to \mathbb{R}^2$, $[T]_\beta^\gamma \in M_{2\times 3}$

② $T(1,0,0,0,0,0) = (2,0,0,0)$

$T(0,1,0,0,0,0) = (-1,1,0,0)$

$T(0,0,1,0,0,0) = (0,1,0,0)$

$[T]_\beta^\gamma = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$T: \mathbb{R}^6 \to \mathbb{R}^4$, $[T]_\beta^\gamma \in M_{4\times 6}$

訂正 $[T]_\beta^\gamma = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

③ $T(1,0) = (2,3,1)$

$T(0,1) = (-1,4,0)$

$[T]_\beta^\gamma = \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}$

④ β = standard ordered bases for $\mathbb{R}^n$, $\beta = \{\beta_1, \beta_2, \beta_3, \ldots, \beta_n\}$

$T(\beta_1) = \beta_n$

$T(\beta_2) = \beta_{n-1}$

$\vdots$

$[T]_\beta = \delta_{i,n+1-j} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & & \iddots & & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$